Instagram Analysis

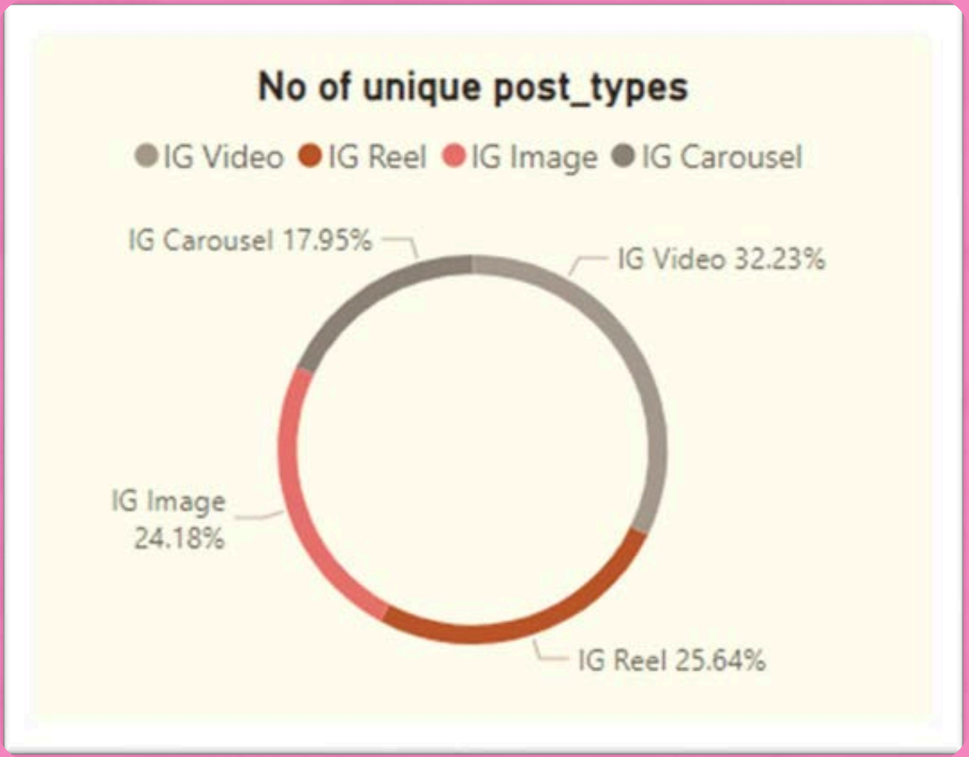
SQL

By Rama Gopala Krishna Masani

1.Unique post types are found in the 'fact content' table

```sql
SELECT
    DISTINCT (post_type) AS Unique_Post_Count
FROM
     fact_content;
```

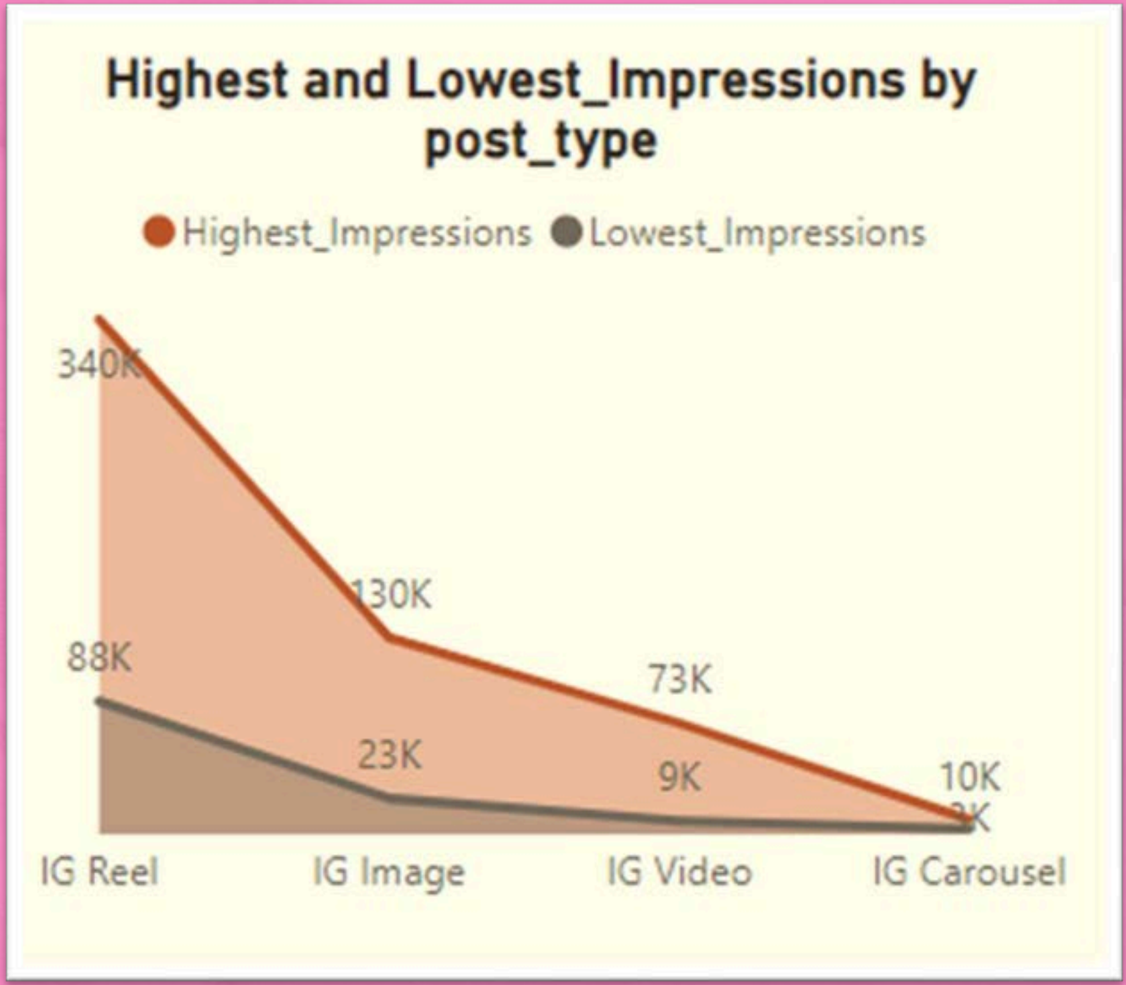
	Unique_Post_Count
▶	IG Image
	IG Reel
	IG Carousel
	IG Video



2The highest and lowest recorded impressions for each post type

```sql
SELECT
    post_type,
    MAX(impressions) AS Highest,
    MIN(impressions) AS Lowest
FROM
    fact_content
GROUP BY post_type;
```

post_type	Highest	Lowest
IG Image	129694	23367
IG Reel	339708	87570
IG Carousel	9677	3264
IG Video	73321	8741



```sql
SELECT
    dd.month_name, dd.weekday_or_weekend,fc.*
FROM
    fact_content fc
        JOIN
    dim_dates dd ON fc.date = dd.date
WHERE
    dd.month_name IN ('March' , 'April')
        AND dd.weekday_or_weekend = 'Weekend';
```

3.Filter all the posts published on the weekend in March and April and export them to a separate CSV file.

month_name	weekday_or_weekend	date	post_category	post_type	video_duration	carousel_item_count	impressions	reach	shares	follows	likes	comments	saves
March	Weekend	03/04/2023	Earphone	IG Video	291	0	12265	3668	69	92	327	7	18
March	Weekend	03/05/2023	Smartwatch	IG Image	0	0	62770	18001	273	360	1194	28	76
March	Weekend	03/11/2023	Mobile	IG Carousel	0	3	5899	1093	45	12	53	0	6
March	Weekend	03/12/2023	Laptop	IG Image	0	0	79416	23474	327	259	1235	69	204
March	Weekend	03/18/2023	Mobile	IG Carousel	0	3	9157	2254	67	58	55	6	15
March	Weekend	03/19/2023	Smartwatch	IG Carousel	0	3	4146	1079	42	17	43	1	6
March	Weekend	03/25/2023	Earphone	IG Reel	22	0	132284	66721	1093	1482	3622	83	695
March	Weekend	03/26/2023	Mobile	IG Image	0	0	63425	26113	435	336	1994	68	179
April	Weekend	04/01/2023	Mobile	IG Carousel	0	3	4549	1052	27	18	35	1	6
April	Weekend	04/02/2023	Earphone	IG Video	163	0	54672	16126	172	182	938	22	81
April	Weekend	04/08/2023	Other Gadgets	IG Video	258	0	37955	12663	204	164	753	31	63
April	Weekend	04/09/2023	Mobile	IG Image	0	0	52278	14438	271	167	1393	36	44
April	Weekend	04/15/2023	Laptop	IG Reel	30	0	123270	39850	296	1486	3926	101	1139
April	Weekend	04/16/2023	Other Gadgets	IG Reel	29	0	115701	66829	937	929	5749	94	658
April	Weekend	04/22/2023	Laptop	IG Video	172	0	33604	14682	255	349	1038	22	73
April	Weekend	04/23/2023	Earphone	IG Video	229	0	36973	13629	224	244	929	30	68
April	Weekend	04/29/2023	Earphone	IG Video	206	0	43526	11799	134	138	646	12	59
April	Weekend	04/30/2023	Mobile	IG Reel	59	0	185017	63990	1010	2238	6039	94	330

4.Create a report to get the statistics for the account. The final output includes the following fields:

- month_name
- total_profile_visits
- total_new_followers

```sql
SELECT
    dd.month_name,
    SUM(fa.profile_visits) AS total_profile_visits,
    SUM(fa.new_followers) AS total_new_followers
FROM
    fact_account fa
        JOIN
    dim_dates dd ON fa.date = dd.date
GROUP BY dd.month_name;
```

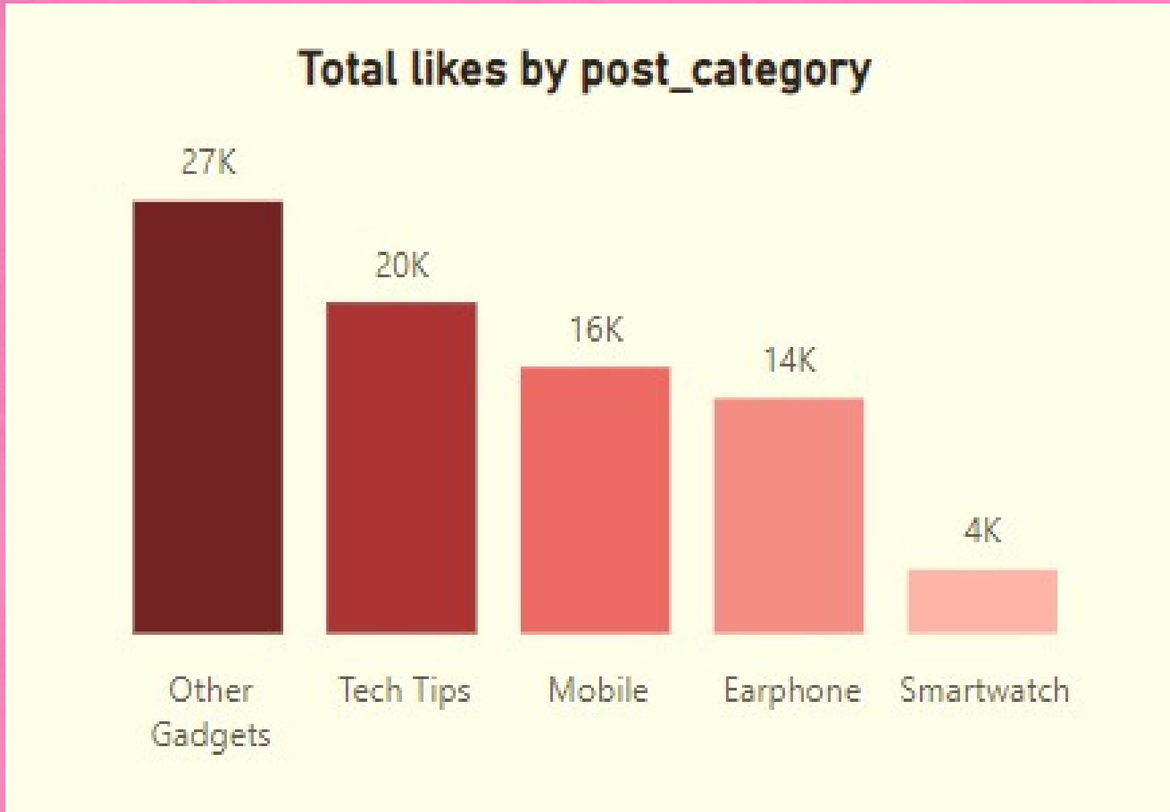
month_name	total_profile_visits	total_new_followers
January	26512	17053
February	20628	15254
March	23132	18285
April	29852	21799
May	106571	66984
June	103350	76942
July	54352	33302
August	42094	24371
September	41522	28523

5.Write a CTE that calculates the total number of 'likes' for each 'post category' during the month of 'July' and subsequently, arrange the 'post category' values in descending order according to their total likes.

```sql
WITH LikesByCategory AS (
    SELECT post_category,
           SUM(likes) AS total_likes
    FROM fact_content
    WHERE MONTHNAME(date) = 'July'
    GROUP BY post_category
)
SELECT *
FROM LikesByCategory
ORDER BY total_likes DESC;

SELECT * FROM gdb0120.likes_by_category_july;
```

post_category	total_likes
Other Gadgets	26519
Tech Tips	20296
Mobile	16338
Earphone	14435
Smartwatch	3918



6. Create a report that displays the unique post_category names alongside their respective counts for each month. The output should have three columns:

- month_name
- post_category_names
- post_category_count

```sql
SELECT
    dd.month_name,
    GROUP_CONCAT(DISTINCT fc.post_category
        ORDER BY fc.post_category) AS post_category_names,
    COUNT(DISTINCT fc.post_category) AS post_category_count
FROM
    fact_content fc
        JOIN
    dim_dates dd ON fc.date = dd.date
GROUP BY dd.month_name;
```

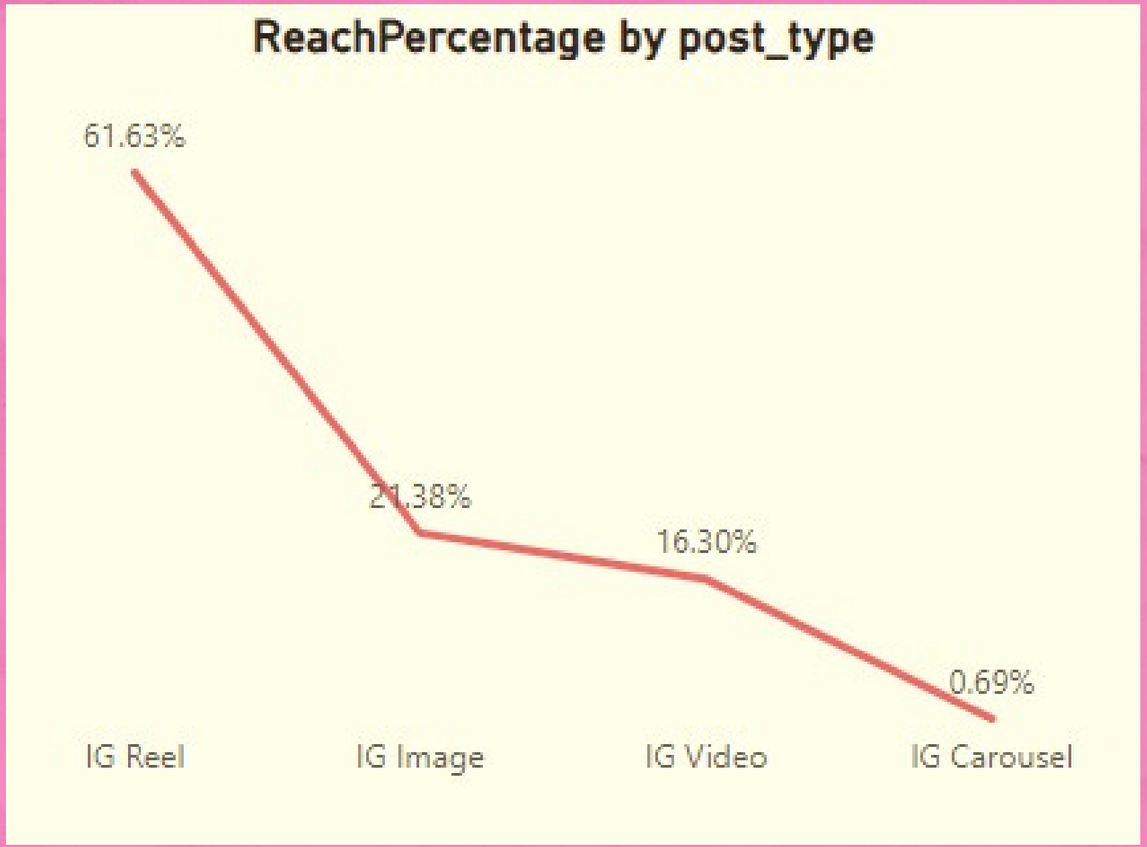
month_name	post_category_names	post_category_count
January	Earphone, Mobile, Smartwatch	3
February	Earphone, Laptop, Mobile, Smartwatch	4
March	Earphone, Laptop, Mobile, Smartwatch	4
April	Earphone, Laptop, Mobile, Other Gadgets, Smartwatch	5
May	Earphone, Laptop, Mobile, Other Gadgets, Smartwatch, Tech Tips	6
June	Mobile, Other Gadgets, Smartwatch, Tech Tips	4
July	Earphone, Mobile, Other Gadgets, Smartwatch, Tech Tips	5
August	Earphone, Mobile, Other Gadgets, Smartwatch, Tech Tips	5
September	Mobile, Other Gadgets, Smartwatch, Tech Tips	4

7. What is the percentage breakdown of total reach by post type? The final output includes the following fields:

- post_type
- total_reach
- reach_percentage

```sql
SELECT
    post_type,
    SUM(reach) AS total_reach,
    ROUND(SUM(reach) * 100.0 / (SELECT
                    SUM(reach)
                FROM
                    fact_content),
            2) AS reach_percentage
FROM
    fact_content
GROUP BY post_type;
```

post_type	total_reach	reach_percentage
IG Reel	5379091	61.63
IG Image	1866381	21.38
IG Video	1422300	16.30
IG Carousel	60465	0.69

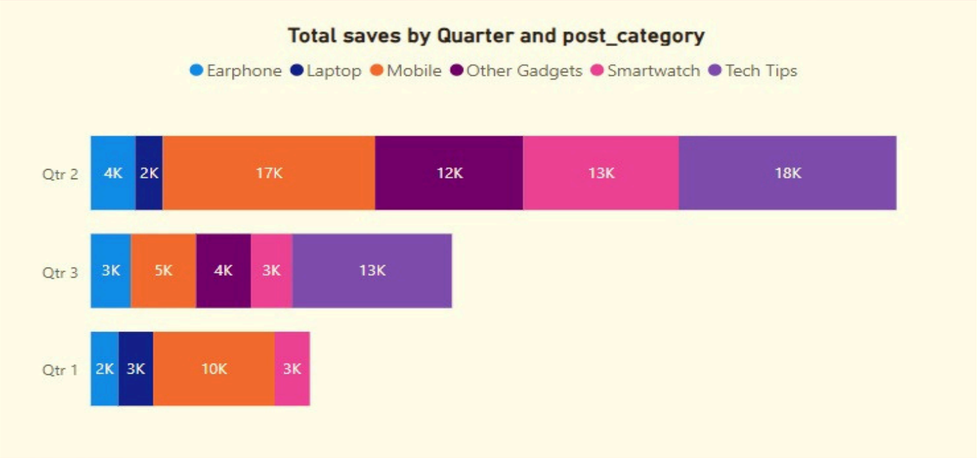


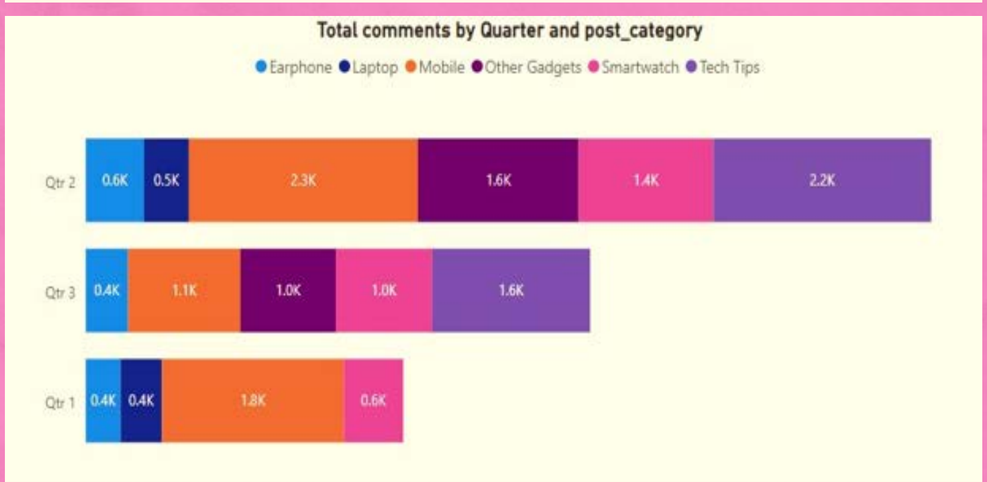
Create a report that includes the quarter, total comments, and total saves recorded for each post category. Assign the following quarter groupings:

(January, February, March) → “Q1”
(April, May, June) → “Q2”
(July, August, September) → “Q3”

```sql
SELECT
    fc.post_category,
    CASE
        WHEN dd.month_name IN ('January' , 'February', 'March') THEN 'Q1'
        WHEN dd.month_name IN ('April' , 'May', 'June') THEN 'Q2'
        WHEN dd.month_name IN ('July' , 'August', 'September') THEN 'Q3'
    END AS quarter,
    SUM(fc.comments) AS total_comments,
    SUM(fc.saves) AS total_saves
FROM
    fact_content fc JOIN dim_dates dd ON fc.date = dd.date
WHERE
    dd.month_name IN ('January' , 'February','March','April','May','June','July','August','September')
GROUP BY fc.post_category , CASE
    WHEN dd.month_name IN ('January' , 'February', 'March') THEN 'Q1'
    WHEN dd.month_name IN ('April' , 'May', 'June') THEN 'Q2'
    WHEN dd.month_name IN ('July' , 'August', 'September') THEN 'Q3'
END;
```

post_category	quarter	total_comments	total_saves
Mobile	Q1	1836	9843
Smartwatch	Q1	600	2860
Earphone	Q1	351	2230
Laptop	Q1	418	2837
Mobile	Q2	2313	17207
Earphone	Q2	589	3602
Smartwatch	Q2	1358	12581
Other Gadgets	Q2	1622	12041
Laptop	Q2	452	2248
Tech Tips	Q2	2201	17649
Other Gadgets	Q3	964	4457
Smartwatch	Q3	971	3326
Earphone	Q3	427	3247
Tech Tips	Q3	1596	12976
Mobile	Q3	1134	5285





List the top three dates in each month with the highest number of new followers. The final output should include the following columns:

- month
- date
- new_followers

```sql
SELECT
    month_name,
    date,
    new_followers
FROM (
    SELECT
        dd.month_name,
        dd.date,
        fa.new_followers,
        ROW_NUMBER() OVER (PARTITION BY dd.month_name ORDER BY fa.new_followers DESC) AS rn
    FROM
        fact_account fa
    JOIN
        dim_dates dd ON fa.date = dd.date
) ranked
WHERE rn <= 3
ORDER BY month_name, new_followers DESC;
```

month	date	new_followers
1	2023-01-30	3186
1	2023-01-03	2959
1	2023-01-23	1003
2	2023-02-01	4106
2	2023-02-24	2383
2	2023-02-02	1989
3	2023-03-21	5421
3	2023-03-28	2513
3	2023-03-25	2356
4	2023-04-25	3736
4	2023-04-30	2753
4	2023-04-06	2500
5	2023-05-08	8872
5	2023-05-20	6169
5	2023-05-12	6051
6	2023-06-30	8804
6	2023-06-03	8802
6	2023-06-21	7033
7	2023-07-08	3716
7	2023-07-15	3364
7	2023-07-28	2344

10. Create a stored procedure that takes the 'Week_no' as input and generates a report displaying the total shares for each 'Post_type'. The output of the procedure should consist of two columns:

- post_type
- total_shares

```sql
CREATE DEFINER=`root`@`localhost` PROCEDURE `GetTotalSharesByPostType`(IN input_week_no VARCHAR(10))
BEGIN
    SELECT
        fc.post_type,
        SUM(fc.shares) AS total_shares
    FROM
        fact_content AS fc
    JOIN
        dim_dates AS dd ON fc.date = dd.date
    WHERE
        dd.week_no = input_week_no
    GROUP BY
        fc.post_type
    ORDER BY
        total_shares DESC;
END;
```

post_type	total_shares
IG Image	861
IG Video	410
IG Carousel	42

```sql
call gdb0120.GetTotalSharesByPostType('W2');
```

Recommendations

1.TrackEngagement Metrics Regularly:

- **Likes, Comments, Shares, and Saves: Monitor these to understand how users interact** with your posts. Higher engagement means your content resonates well with your audience.
- **Engagement Rate**: Keep track of this by dividing total interactions by impressions or reach. It gives you a better sense of content performance relative to audience size.

2. Measure Content Reach and Impressions:

- **Reach**: Track the number of unique users who see your posts to understand how far your content is spreading.
- **Impressions:** Monitor this to see how many times your content has been displayed, which is useful for gauging visibility.

3. Content Diversity:

- Post a variety of content (images, videos, carousels, reels) to keep your audience engaged. Track which format performs best.

Thank You